SMRFORGE

VERIFIABLE SMR COMPUTATIONAL ENGINEERING

smrforge.io · Open-core · Community open source · Pro in beta

SMR DESIGN · ANALYSIS · OPTIMIZATION

Reactor physics your reviewers can reproduce — not just believe.

A full screening reactor-physics engine for Small Modular Reactors where every result ships as a fingerprinted, independently reproducible evidence bundle — and is labeled, honestly, for how far to trust it.

`git clone` → `pip install -e .` → `python demo/walkthrough.py` → a screening k-eff you verify in a minute

SMRFORGE.EVIDENCE_ENVELOPE.V1	◆ SEALED
analysis	smr-core-screening
k_eff	1.29841
fidelity	screening
sha-256	a3f1c8…9e0b
replay	reproduce · bit-exact

```
$ smrforge verify ./bundle
✓ sealed · schema ok · unedited · exit 0
```

THE DIFFERENTIATOR — BUILT FOR THE PERSON WHO DOESN'T TRUST YOU

01 · FINGERPRINT

Evidence envelopes

Every output is a schema-registered bundle sealed by a SHA-256 manifest of its inputs and result — any edit breaks the seal, and it is `ed25519`-signable (a demo identity today, production key on the roadmap).

02 · REPLAY

Bit-exact — or a sealed tolerance

`smrforge reproduce <bundle>` re-runs the open engine on the bundle's own recorded inputs and re-derives the result core. A deterministic core re-derives bit-for-bit (pinned data library, seed, and code version); a stochastic Monte Carlo result re-derives within a statistical tolerance sealed in the record — so it can't be quietly widened.

03 · VERIFY

`exit 0` = it holds

A standalone, offline check against the open **Evidence Envelope** schema — a third party confirms your record's integrity (unedited, schema-valid) or refutes it in seconds, no license needed. Each result also pins a fingerprint of the data it read, or is honestly marked `unpinned`.

ONE REPRODUCIBLE MODEL, A FULL SCREENING REACTOR-PHYSICS STACK — AN LWR-SMR SCREENING PRESET TODAY; HTGR, MSR & MICROREACTOR ARCHETYPES ON THE ROADMAP

NEUTRONICS	Full-core nodal & NEM · CE Monte Carlo & S_n transport	KINETICS	Point-kinetics · reactivity coefficients	FUEL CYCLE	Burnup & depletion · Xe/Sm poisoning
SAFETY	Rod-worth & shutdown-margin · pin-power	UNCERTAINTY	Data-uncertainty · sensitivity	NUCLEAR DATA	In-tree ENDF/B · self-shielding, TSL
GEOMETRY	Cartesian cores, full-core CMFD & R-Z — free · hex on the open-core roadmap	EVIDENCE	Sealed bundles · offline verify · data & derivation provenance	INTEROP	OpenMC round-trip · Serpent/MCNP converters on engagement

HONEST BY DEFAULT · OPEN-CORE

WHAT IT GIVES YOU

- ✓ A `fidelity` label + plausibility band on every result
- ✓ Machine-readable `caveats` that travel with the output
- ✓ One-command escalation to OpenMC (free) for a reference-grade k-eff

WHAT IT WON'T PRETEND

- Screening k-eff is a trend, not a validated absolute
- The open core is honestly screening-grade, not licensing sign-off
- Bundles seal under a published demo identity — a check proves integrity, not origin
- Nothing is tuned to hit a benchmark it was measured against

Community `smrforge · MIT`

The full screening engine — prove fit in the open.

- Full screening stack (nodal + NEM, CE-MC, S_n, burnup, kinetics) + evidence spine & reviewer-kit
- An LWR-SMR screening preset → a runnable, baseline-checked model via the Python API (HTGR/MSR/micro archetypes on the roadmap)
- One-step OpenMC escalation, free (Serpent/MCNP converters on engagement)
- Benchmarks, Python API; CLI ships `verify`, `reproduce`, `license` & `--version`

Pro `smrforge_pro · FSL-1.1-MIT` COMMERCIAL

The assurance / regulatory layer for submissions.

- The signed `assess` deliverable — a real reference solve + a tech-spec shutdown-margin gate in one bundle, with a PASS/FAIL verdict against the safety floor
- Coverage (USL / c_k trending) · freshness watch · signed, shareable evidence reports
- Independent review is **free & open** today — the reviewer kit re-runs the pinned engine and verifies with the separate `evidence-verifier`; a hosted reviewer console + requirements-to-evidence traceability & SRP mapping are on the roadmap
- Deck converters for your own licensed Serpent/MCNP, on engagement *(physics fidelity is free open-core — CMFD & R-Z ship free; hex is on the open-core roadmap)*

```
# Screening k-eff + live verify, from a checkout: $ python demo/walkthrough.py # → computes k, verifies it with a standalone checker, then catches a signed lie
```

Open standard & verifier →

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